



Stage 1: Business Discovery Report

Business Process Mapping & Domain Foundations

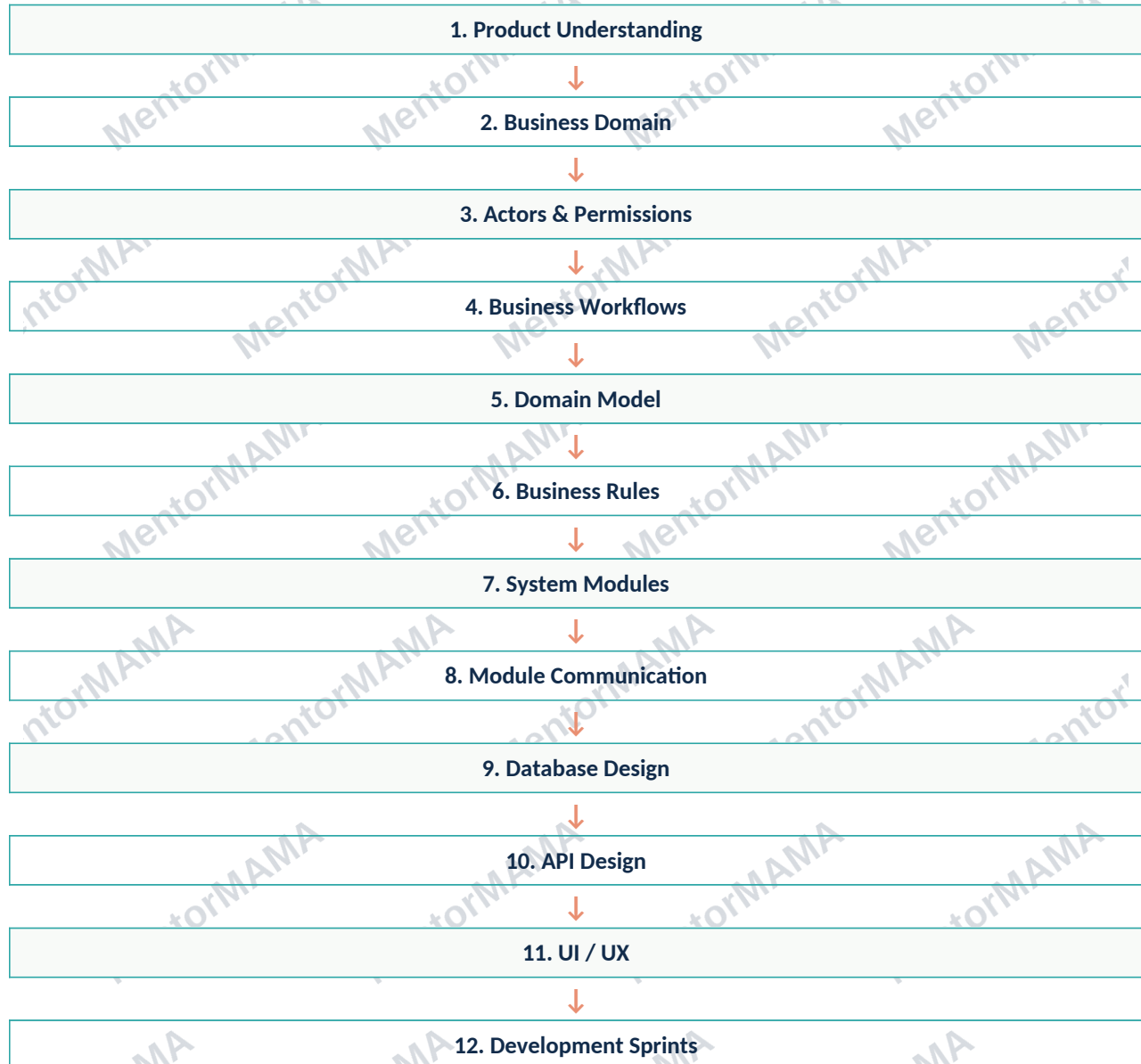
Understanding the real-world mentorship ecosystem before a single table, screen, or endpoint is designed

Field	Detail
Product	MentorMAMA — Digital Mentorship Toolkit for Safer Midwifery Clinical Placement
Prepared by	Sinaps Technology — Bouric Okwaro Enos (Ric), Co-Founder & Team Lead
Document type	Stage 1 of the design sequence: Business Discovery, preceding Domain Modelling, Business Rules, Module Architecture, Database Architecture, and beyond
Version	1.0
Date	July 2026

Better mentorship. Better outcomes.

1. Purpose of This Stage

A database is a persistence mechanism for the business; it should not define the business. Jumping into tables or screens first is one of the most common reasons systems become difficult to evolve. This document instead follows a strict order of design, where each stage informs the next, and the database does not appear until step 9 of 12:



This document delivers Stages 1–2 in full and lays the foundation — actors, ecosystem, process, nouns, verbs — for the Domain Model that follows in Stage 3.

2. What MentorMAMA Actually Is

MentorMAMA is a governance layer over a relationship that already exists.

A university sends a student into a hospital ward it does not control, for a period of weeks, expecting that a stranger employed by that hospital will supervise, teach, and vouch for that student's clinical competence. That relationship today runs on goodwill, paper checklists, and whoever happens to be on shift.

MentorMAMA does not invent mentorship — it makes an already mandatory, already happening relationship visible, accountable, and measurable to the two institutions that share the outcome but have no shared system today.

The real customers are not the student, and not even the mentor. They are the University — accountable for producing competent nurses — and the Hospital — accountable for what happens on its wards. Everything else in the system is instrumentation on top of that relationship.

2.1 The Problem That Exists Without It

- A university has no visibility into whether a student was actually mentored this week, or just physically present.
- A hospital has no structured record of which of its staff are mentoring, how much time it costs them, or whether they are any good at it.
- Neither side has an early-warning signal when a placement is going badly — it surfaces only after something has already gone wrong.
- Mentorship quality is undocumented, so it cannot be improved, credited through CPD, or defended if something is disputed later.

MentorMAMA's job is to close that visibility gap between two organizations that jointly own an outcome but do not share a system.

3. The Real-World Process, In Full

Software should digitize and improve this existing process — not invent a new one. The lifecycle runs in three phases: institutional setup (once per term), the placement itself (repeats per student), and closure & reporting.

3.1 Pre-Placement — Institutional Setup

1. University and Hospital have, or negotiate, a placement partnership.
2. University Coordinator builds a cohort for the term.
3. Coordinator requests placement capacity from one or more hospitals.
4. Hospital confirms capacity per ward, via Nurse Manager or Program Administrator.
5. Coordinator allocates named students to named hospitals and wards.
6. Nurse Manager identifies which of their staff will mentor this cohort.
7. Mentors who have not yet done so complete mentorship training.

3.2 During Placement — Repeats Per Student

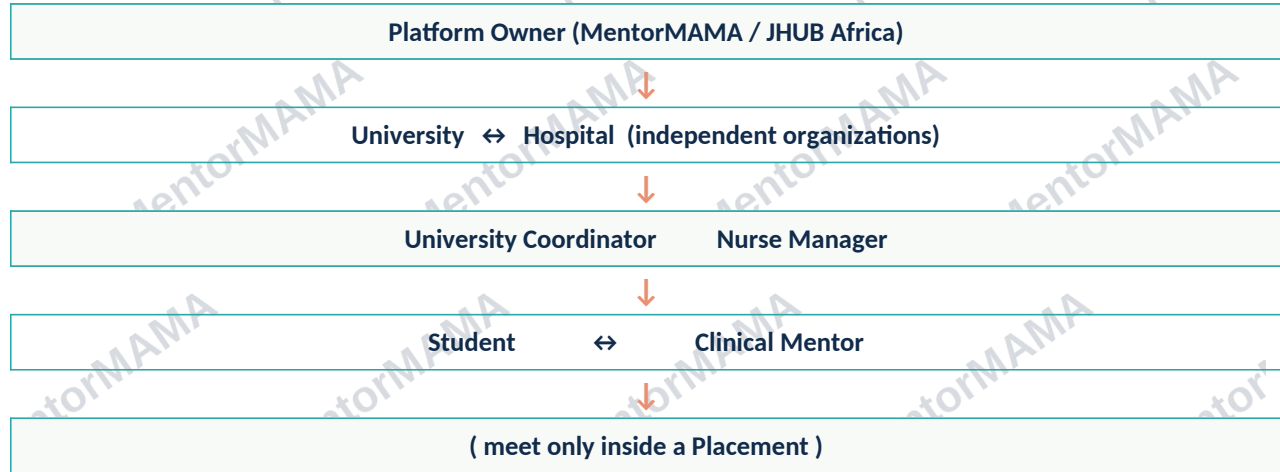
8. Student arrives at the hospital gate.
9. Student reports to the maternity unit.
10. Nurse Manager verifies identity and assigns or confirms a mentor.
11. Mentor conducts labour ward induction: safety, workflow, reporting lines, respectful maternity care.
12. Mentor supervises, teaches, and observes daily/weekly.
13. Mentor logs each mentorship interaction.
14. Student periodically reports confidence and flags concerns.
15. If something goes wrong: an escalation is raised and the Nurse Manager or Coordinator intervenes.
16. Mentor reviews pending follow-ups and adjusts supervision.

3.3 Post-Placement — Per Student, Then Per Cohort

17. Student completes an exit survey and final confidence self-rating.
18. Mentor's involvement with this student closes.
19. Nurse Manager reviews the ward-level placement summary.
20. University Coordinator receives the placement completion report.
21. Where applicable, the mentor's training/CPD completion is reviewed and certified.
22. Program Administrator rolls facility and cohort data into program-level reporting for JHUB Africa.

4. The Organizational Ecosystem

MentorMAMA is not owned by a hospital, and it is not owned by a university. It is a collaborative platform connecting independent organizations that meet at exactly one shared point: the Placement.



A Nurse Manager does not outrank a University Coordinator. Each has total authority inside their own institution and zero authority inside the other. This is not a chain of command — it is a set of non-overlapping jurisdictions that meet at one shared object: the Placement.

5. Actor Analysis

Actor	Works For / Belongs To	Responsibilities
Student	University (belongs to, not employed)	Attend placement, learn, complete orientation, meet mentor, attend sessions, submit feedback, report concerns. Never manages anyone — the consumer of mentorship.
Clinical Mentor	Hospital (employee: midwife, senior nurse, clinical instructor)	Guide students, complete induction, train students, assess progress, log mentorship sessions, support students.
Nurse Manager (Ward In-Charge)	Hospital (operational manager of the ward)	Oversee the maternity ward, assign mentors, monitor mentorship, resolve issues, ensure supervision, review reports. Manages the ward, not every student.
University Coordinator	University	Upload students, create placement cohorts, allocate students to hospitals, monitor progress, receive reports, communicate with hospitals. Does not work for hospitals.
Program Administrator	Platform / Program (JHUB Africa)	Configures the software, not healthcare: creates hospitals, institutions, wards; manages users & permissions; uploads content; system settings; reports. Functionally like a Workspace admin.
Notification Service	Not human — system actor	Dispatches email (now) and SMS/WhatsApp (later) on domain events such as a placement being assigned.

5.1 Platform Owner vs. Program Administrator — Design Decision

These are conceptually distinct: owning the platform, licensing, and cross-program governance is a different responsibility from running day-to-day configuration of one program's facilities and users. For this pilot, they are the same small group of people.

Decision: keep Program Administrator as the one operational role, scoped by a program identifier, with a small number of accounts flagged as platform-level, rather than building two fully separate role hierarchies now. This preserves a future multi-tenancy path without inventing a role nobody uses yet.

5.2 CPD Reviewer — Demoted From the MVP Actor Set

This role does not touch the placement workflow: it does not create placements, onboard students, log mentorship sessions, manage wards, or resolve placement issues. It only reviews mentor learning content and, where applicable, confirms completion for CPD purposes.

Decision: CPD Reviewer stays in the domain vocabulary for Phase 2, but is out of the MVP authorization model. The concept note itself defers regulator integration to a later phase.

6. Domain Nouns and Verbs

6.1 Nouns — Candidate Domain Entities

Category	Nouns
Organizational (own data, don't log in)	Platform / Program, Institution, Hospital, Ward
People (log in, act on behalf of an org)	Program Administrator, University Coordinator, Nurse Manager, Clinical Mentor, Student
Workflow / Relationship (the actual business events)	Cohort, Placement, Induction Checklist, Mentorship Session, Feedback, Escalation
Capability / Content	Training Module, Module Completion, Certificate (future)
Infrastructure	Notification, Report, Dashboard

6.2 Verbs — Candidate Business Actions

- Negotiate placement capacity (Coordinator ↔ Hospital)
- Build a cohort (Coordinator)
- Allocate student to hospital/ward (Coordinator)
- Assign / reassign mentor to student (Nurse Manager)
- Start / End / Pause a placement (Nurse Manager, system)
- Complete induction (Mentor)
- Log a mentorship session (Mentor)
- Submit feedback (Student)
- Raise an escalation (Student, Mentor)
- Review / close an escalation (Nurse Manager, Coordinator)
- Complete a training module (Mentor)
- Confirm module completion (Program Admin / future CPD)
- Generate a report / dashboard (Coordinator, Nurse Manager, Admin)

7. Placement as the Aggregate Root

Validating which noun owns which verb is what confirms the center of the domain model.

Noun	Owns	Role in the Model
Placement	Start, End, Pause, Reassign Mentor	Aggregate root — anchors Induction Checklist, Mentorship Session, Feedback, and Escalation. None of those four objects mean anything without an active Placement.
Cohort	Build, Close, Allocate Student	Sits one level above Placement — the batch a Placement is drawn from.
Training Module	Publish, Complete, Confirm Completion	Deliberately independent of Placement — a mentor's training is facility-independent and outlives any single student relationship.
Escalation	Raise, Review, Resolve	References a Placement but has its own lifecycle and its own restricted visibility rules.

Conclusion: Placement is confirmed as the aggregate root of the domain model. Every workflow object that happens between a specific student and mentor, in a specific ward, in a specific time window, is a child of Placement.

8. Data Ownership

This is the question underneath “who has authority over whom” — and it is what the authorization model in Stage 3 will be built on.

Data	Owned By	Who Else Can Read It
Student master record (name, reg. no, course)	University	Hospital sees only students placed with them
Mentor master record (cadre, experience)	Hospital	University does not see hospital staff records at all
Cohort	University	Hospital sees only the slice allocated to them
Placement	Jointly created, platform-owned	Both University and Hospital see their side of it
Induction Checklist, Mentorship Session	Hospital (created by their mentor)	University sees aggregate/summary, not raw entries, unless escalated
Feedback	Student (submitted)	Visibility governed by policy — open decision (Section 9)
Escalation	Hospital (safeguarding-sensitive)	University sees it only above a severity threshold or on resolution failure
Facility / Ward configuration	Hospital	Program Admin can configure on their behalf during onboarding
Program-level reports	Program Administrator	Rolled up from both sides; shown to funders / partner programs

9. Design Decision: The Placement Capacity Question

Open question: should the pre-placement capacity negotiation between Coordinator and Hospital become an in-app workflow?

9.1 Decision

Keep it out of the MVP as a workflow. Model for it in the data, but don't build UI or a state machine around it yet.

9.2 Reasoning

- At pilot scale — one or two maternity units already linked to JKUAT placement sites — this negotiation is a conversation the Nurse Manager and Coordinator have once, before either logs into the system, since a partnership agreement necessarily precedes any placement activity.
- The real shape of that negotiation — per-ward or per-shift capacity, partial confirmation, mid-cohort renegotiation — is not yet known. Building a state machine on guesses risks exactly the expensive rework this design discipline exists to avoid.
- The Program Administrator is already creating the Cohort by hand during pilot setup regardless, so a workflow isn't solving a problem the pilot actually has yet.

9.3 Resolution

Cohort carries a simple `planned_capacity` field, filled in by the Program Administrator from whatever the offline conversation produced. No request/confirm workflow, no capacity state machine, in the MVP.

Revisit trigger: this becomes a real feature once MentorMAMA runs more than two or three hospital partners, or once a Coordinator is juggling multiple institutions competing for placement slots at the same hospital.

10. Stage 1 Outcome

10.1 The Five Questions, Answered

Question	Answer
What is MentorMAMA?	A shared visibility and accountability layer between a University and a Hospital over a mentorship relationship neither institution can fully see today.
What business problem does it solve?	Two institutions jointly own an outcome with no shared system, no early-warning signal, and no record of mentorship quality.
What happens in the real world today?	A three-phase lifecycle — institutional setup, the placement itself, and closure/reporting — running almost entirely on informal coordination.
Who participates?	Two independent organizations (University, Hospital) each with their own internal hierarchy, meeting at exactly one shared object: the Placement, under a Platform layer that configures and monitors the program.
What are the core business objects and actions?	Placement is the aggregate root. Cohort and Training Module are independently-owned adjacent objects. Induction Checklist, Mentorship Session, Feedback, and Escalation are all children of an active Placement.

10.2 What Stage 2 Will Build On This

Stage 2 — Domain Modelling — will take the nouns, verbs, and ownership boundaries confirmed here and turn them into classes, attributes, and relationships: validating Placement as the center of the system and defining exactly what each object is responsible for.



End of Stage 1: Business Discovery